

ULTRA-SAFE CENTRAL CAPSULE

DOCUMENT 00 — COMPREHENSIVE ENGINEERING BASELINE

Version 3.0 — Corrected, Controlled and Verification-Gated Baseline

Document status	Controlled engineering baseline — not a certification approval
Primary purpose	System-level definition, requirements control, verification planning and IP traceability
Baseline geometry	5.80 m external diameter × 3.80 m overall height
Occupancy	7 persons: pilot, co-pilot, 5 additional occupants
Mass ceiling	22,000 kg maximum operational mass / hard design ceiling
Safety architecture	Six-layer survival architecture
Landing architecture	8 landing legs with progressive energy absorption
Recovery architecture	8 primary + 8 auxiliary/stabilizer parachute concept; final sizing verification required
Life support target	72 h for 7 occupants
Variants	Helicopter, fixed-wing/jet, maritime/ship; derivative rules controlled herein

Engineering integrity statement: This document intentionally distinguishes fixed baseline values, engineering requirements, assumptions, derived targets, and verification gates. No performance value is treated as demonstrated unless supported by controlled analysis, test evidence, or an approved qualification report.

Revision purpose: consolidate the original Version 1.6 baseline while correcting internal accounting, requirement definitions, configuration control, safety independence claims, parachute sizing basis, impact-energy accounting, host compatibility, flotation, life-support, electrical/software common-cause risk, verification, and engineering-to-IP traceability.

DOCUMENT CONTENTS

- 1. Document Authority, Scope and Status
- 2. Mission, Design Philosophy and Survival Envelope
- 3. System Definition and Functional Decomposition
- 4. Top-Level Requirements
- 5. Geometry, Packaging and Occupancy
- 6. Mass Budget and Mass-Properties Control
- 7. Centre of Gravity, Inertia and Configuration States
- 8. Primary Structure — CFRP Roll-Cage and Continuous Floor
- 9. Six-Layer Safety Architecture — Detailed Engineering Requirements
- 10. Emergency Detection, Decision Logic and Timing
- 11. Host Separation and Clean-Break Interface
- 12. Parachute Recovery System — Required Sizing Basis
- 13. Landing Legs and Impact Energy Management
- 14. Occupant Protection, Seats, Restraints and Egress
- 15. Life Support, Atmosphere and Thermal Control
- 16. Fire, Smoke and Thermal Protection
- 17. Flotation, Stability, Flooding and Self-Righting
- 18. Electrical Power, Software and Common-Cause Control
- 19. Pyrotechnic, Thermal-Battery, Pressure-Vessel and Stored-Energy Safety
- 20. Environmental Qualification
- 21. Host-Platform Interface Control
- 22. Variant and Family Architecture
- 23. Scalability Rules
- 24. Verification, Validation and Test Programme
- 25. Mandatory Safety Analyses
- 26. Human Factors and Operational Procedures
- 27. Requirements Traceability and Evidence Database
- 28. Configuration Management
- 29. Engineering-to-IP Traceability
- 30. Engineering Maturity Gates
- 31. Mandatory Engineering Work Packages Before Performance Claims
- 32. Controlled Baseline Summary
- 33. Corrected Deficiency Register — Version 1.6 to Version 3.0
- 34. Final Engineering Position

1. Document Authority, Scope and Status

Document 00 is the controlling system-level engineering baseline for the Ultra-Safe Central Capsule concept. All detailed design, analysis, simulation, testing, certification planning, patent extraction and licensing work shall trace to this baseline or to an approved lower-level controlled document.

The baseline defines the intended architecture and the values that are presently locked at system level. It does not claim that every subsystem has reached detailed-design maturity. Where a quantity requires engineering closure, this document defines the required closure method and acceptance evidence instead of inventing a result.

- **Baseline:** Version 3.0 supersedes the internal engineering logic of Version 1.6 while retaining its core system concept and locked geometry unless explicitly changed here.
- **Requirement:** A mandatory technical statement that must be satisfied and verified.
- **Target:** A design objective that becomes a requirement when formally baselined by configuration control.
- **Assumption:** A provisional input that must be confirmed before dependent claims are released.
- **Derived value:** A calculated result that must be reproducible from controlled inputs.
- **Verification evidence:** Analysis, inspection, demonstration, test, or certified data appropriate to the requirement.

Important limitation: No engineering document can honestly guarantee “zero defects” before the design is analyzed, built, tested and qualified. Version 3.0 therefore removes false certainty and makes unresolved items explicit, controlled and verifiable.

2. Mission, Design Philosophy and Survival Envelope

The mission is to provide a survivable, self-contained occupant capsule capable of separating from a host platform or surviving a severe host accident, managing descent or impact energy, protecting occupants, and supporting post-event survival and recovery.

The design philosophy is layered protection rather than reliance on a single rescue mechanism. The architecture combines emergency detection/decision logic, multiple separation initiation paths, controlled separation hardware, parachute recovery, direct-impact energy management, and post-event survival/recovery functions.

- The capsule shall be designed around credible failure cases rather than nominal operation alone.
- Every claimed redundancy shall identify its common power, sensing, software, structure, environment and human dependencies.
- Survival claims shall be stated against defined boundary conditions: mass, attitude, speed, altitude, atmosphere, terrain/water state, occupant configuration and system health.
- Variant-specific aerodynamic and host-interface limits shall not be silently inherited from another vehicle class.

The present document does not establish a single universal “survivable accident” number. Such a number requires the controlled energy, load, injury and structural criteria defined in Sections 12–17 and verified through analysis and testing.

3. System Definition and Functional Decomposition

Function	Baseline definition	Closure evidence
Host attachment	Carries the capsule through normal operation and all specified host loads	Host interface ICD + structural/aero analysis + qualification test
Emergency detection	Detects defined accident/emergency triggers and issues a controlled decision	Sensor FMEA + timing analysis + fault injection
Separation	Creates a clean, controlled separation path from the host	Interface load analysis + hardware-in-loop + separation test
Recovery	Deploys and controls parachute recovery when within the qualified envelope	6-DOF trajectory + CFD/aero + deployment test + failure cases
Impact protection	Absorbs landing/impact energy and limits occupant loads	Nonlinear FEA + drop/impact test + injury criteria
Occupant protection	Restraints, seats, energy absorption and egress	Dynamic test + human factors + egress demonstration
Post-event survival	Atmosphere, thermal, fire, flotation and location functions	Endurance, environmental, flotation and recovery tests
Recovery/location	Transmits location and supports visual/acoustic recovery	System test + RF/antenna qualification + field test

Functional decomposition shall be expanded into a complete product breakdown structure before detailed design release. Every subsystem shall have an owner, interface definition, requirement set, verification method and configuration identifier.

4. Top-Level Requirements

ID	Requirement / controlled value	Verification
REQ-001	External diameter = 5.80 m baseline	Inspection / CAD
REQ-002	Overall height = 3.80 m baseline	Inspection / CAD
REQ-003	Occupancy = 7 persons	Inspection / HF analysis
REQ-004	Maximum operational mass / hard ceiling = 22,000 kg	Mass properties + configuration audit
REQ-005	Primary cage/floor structural safety-factor target ≥ 2.8 , with limit/ultimate basis explicitly defined before substantiation	Analysis + test correlation
REQ-006	Eight landing legs	Inspection + functional test
REQ-007	72 h life-support target for 7 occupants	Mass/energy balance + endurance test
REQ-008	406 MHz COSPAS-SARSAT + GNSS + satellite messaging location architecture	System test
REQ-009	Two side doors + one roof hatch with local structural reinforcement	CAD + structural analysis + egress test
REQ-010	Six-layer survival architecture retained at system level	Safety case + FMEA/FTA/CCA + tests
REQ-011	All safety-critical single-point failures identified and either eliminated, tolerated, or formally accepted	FMEA/FTA/CCA
REQ-012	No performance claim released without a defined verification method and acceptance criterion	Configuration audit

5. Geometry, Packaging and Occupancy

The locked baseline envelope is 5.80 m external diameter and 3.80 m overall height. The stated usable internal volume target of approximately 55–62 m³ shall be treated as a verification target, not a certified fact, until the controlled CAD model defines the internal boundary, occupied volume, equipment volume, passageways and unusable regions.

- Nominal seating: 7 occupants — pilot, co-pilot and five additional occupants.
- Provide controlled clearance zones for seats, restraints, controls, emergency equipment, life-support hardware, parachute deployment hardware and egress paths.
- Two side doors and one roof hatch require local load-path reinforcement without creating an unacceptable structural discontinuity.
- Maintain a controlled digital mock-up for normal, crash, inverted and post-deformation egress conditions.

- Define mass and volume allowances for every equipment item; no “miscellaneous” volume may remain uncontrolled at detailed-design release.

Packaging shall be checked for all configuration states in Section 7, including maximum occupant loading, asymmetric loading, consumable depletion, parachute deployment state and post-separation condition.

6. Mass Budget and Mass-Properties Control

The original component ranges do not mathematically consume the full 22,000 kg ceiling. Their minimum sum is approximately 19,740 kg and their maximum sum is approximately 21,290 kg. The resulting unexplained residual is approximately 2,260 kg at the minimum case and 710 kg at the maximum case. Version 3.0 therefore prohibits treating this residual as an invisible allocation.

Mass item	Original range (kg)	Control rule
Primary structure	6,300–6,600	Controlled design mass + tolerance
Landing legs	2,600–2,800	Controlled design mass + tolerance
Parachute system	1,450–1,600	Includes deployment/recovery hardware; final sizing required
Separation system	950–1,050	All actuators, interfaces, stores and wiring
Energy/electrical	1,850–2,050	Batteries, distribution, conversion, reserves
Life support	1,150–1,250	72 h equipment and consumables
Flotation/location	900–1,000	Buoyancy, recovery and location equipment
Fire/sealing/interiors	1,100–1,200	Protection, liners, seals, interior hardware
Seats/HF	980–1,080	Seats, restraints, human-interface hardware
Occupants	560	Baseline occupant mass allowance
Contingency	1,900–2,100	Must be decomposed; not a substitute for engineering accounting
Unexplained residual to 22,000	710–2,260	Explicit headroom/reserve; not allocated hardware

Mass-control rule: At every design gate, the capsule shall maintain a Current Mass Estimate, Design Mass, Manufacturing Tolerance, Growth Allowance and Remaining Headroom. The 22,000 kg ceiling shall not be exceeded by design evolution. Any remaining headroom shall be visible and numerically reconciled.

7. Centre of Gravity, Inertia and Configuration States

The original concept stated that the centre of gravity should remain within 120 mm of the geometric centre in horizontal and vertical directions. Version 3.0 converts this into a configuration-control requirement rather than a single nominal point.

- Define CG in all three axes relative to a fixed capsule coordinate system.
- Evaluate empty/dry, nominal occupancy, maximum/minimum occupant mass, asymmetric occupant distribution, consumable full/depleted states, parachute armed/deployed states, post-separation configuration, and flotation/recovery configurations.
- Calculate principal moments and products of inertia for each safety-critical configuration.
- Define allowable CG and inertia envelopes for host compatibility, parachute deployment, impact dynamics and self-righting.
- Verify by controlled mass-properties measurement on the article or validated digital mass-properties model.

A single nominal CG value is insufficient for dynamic certification. The controlled envelope shall be used by the parachute, host-interface, impact and stability analyses.

8. Primary Structure — CFRP Roll-Cage and Continuous Floor

The primary structure remains a continuous CFRP roll-cage with titanium nodes and a continuous floor. Primary tube dimensions are retained as a 15–18 cm class baseline, subject to detailed sizing and manufacturing constraints.

- Define laminate stacking sequences, ply materials, thicknesses, fibre directions, cure process and allowable properties.
- Define titanium grade, heat treatment, fasteners/bonding strategy and galvanic isolation.
- Analyze global strength, local buckling, crippling, joint bearing/bypass, pull-through, delamination, impact damage, fatigue and environmental degradation.

- Define load paths from host interface, parachute risers, landing legs, seats, doors, hatch and flotation loads into the cage.
- Use nonlinear structural analysis where geometry, material behaviour, contact, progressive failure or large deformation requires it.
- Correlate analysis with representative component and subassembly tests before full-scale qualification.
- Define manufacturing inspection, NDI/NDT, repair limits and damage-tolerance rules.

The ≥ 2.8 structural safety-factor target shall not be used as a standalone proof. The basis load case, limit load, ultimate load, material allowables, knock-down factors, damage state and failure mode must be defined in the structural substantiation plan.

9. Six-Layer Safety Architecture — Detailed Engineering Requirements

Layer	Function	Independence requirement	Required evidence
1	Primary automatic emergency detection / initiation	Independent sensing and decision path from normal host controls	FTA/FMEA + fault injection
2	Independent backup initiation path	No common single sensor/processor/power dependency with Layer 1	CCA + hardware-in-loop
3	Independent manual/crew initiation	Available when automation is unavailable and human action is feasible	HF + simulator + operational test
4	Independent final/last-resort separation path	Physically and electrically separated where practicable	Separation failure-injection test
5	Direct-impact survival	Does not depend on successful parachute deployment	Impact energy analysis + drop/impact test
6	Post-event survival and recovery	Atmosphere, fire, flotation, location, egress and rescue support	Endurance + environmental + recovery test

The term “independent” shall mean demonstrably independent against defined common-cause failures. Two channels using the same power source, sensor, data bus, software library, connector family, structural path or environmental vulnerability are not automatically independent.

- Perform Functional Hazard Assessment, FMEA/FMECA, Fault Tree Analysis and Common-Cause Analysis.
- Identify latent failures, dormant channels and maintenance-induced common causes.
- Inject representative failures into the detection, initiation, power, communications and actuator paths.
- Establish a quantitative safety case linking layer availability, failure probability and required survival function.

10. Emergency Detection, Decision Logic and Timing

The emergency system shall detect defined accident precursors and execute a deterministic, bounded decision sequence. The system shall not rely on an unspecified “AI decision” or an unbounded software response.

Element	Engineering requirement
Sensors	Use independent modalities as required by hazard analysis: inertial, attitude/rate, air-data/altitude where applicable, host-state and structural/energy indicators.
Decision logic	State-machine or otherwise formally controlled logic with defined thresholds, hysteresis, voting and inhibit conditions.
Timing	Define detection, confirmation, command, actuator initiation, separation and parachute deployment timing budgets.
False activation	Analyze nuisance-trigger probability and operational inhibit/arm logic.
Power loss	Define safe behaviour under total/partial power loss.
Data integrity	Define sensor plausibility, bus fault handling and stale-data detection.
Evidence	Hardware-in-loop, software verification, fault injection and end-to-end timing test.

11. Host Separation and Clean-Break Interface

The separation system shall support the 22,000 kg capsule only when the host platform is specifically designed and substantiated for that mass and the resulting static, dynamic, aerodynamic and separation loads. No generic host compatibility is implied.

- Parallel primary separation actuators with a mechanically independent fallback architecture are retained as the concept basis.
- Cold-gas or equivalent final fallback may be used only after independent hazard and pressure-system qualification.
- Define separation plane geometry, load transfer, actuator stroke, sequencing, interlocks, safe states and residual-energy limits.
- Provide clean-break interfaces for mechanical, electrical, data and fluid services.
- Prevent cable/line snagging, structural interference and recontact after release.
- Perform separation analysis for nominal, off-axis, asymmetric, actuator-failed and host-deformed conditions.
- Qualify the separation interface with representative dynamic testing.

The interface control document shall be jointly controlled by the capsule and each host platform. A capsule design cannot be certified independently of the host loads and separation environment.

12. Parachute Recovery System — Required Sizing Basis

The baseline concept uses 8 primary canopies plus 8 auxiliary/stabilizer canopies, with pilot/extractor chutes. This is an architectural starting point, not a completed parachute design.

For a total mass m and target steady descent speed V , the minimum conceptual drag-area relationship is $A = 2mg/(\rho \cdot C_d \cdot V^2)$. This equation is only a sizing starting point; the actual system must account for deployment dynamics, reefing, line/riser loads, canopy interactions, Mach effects, altitude, temperature, density, opening shock and controllability.

Parameter	Must be defined before performance claim
Mass	22,000 kg maximum and all qualified lower-mass states
Target descent speed	Explicit value with vertical and lateral criteria
Deployment altitude	Minimum/nominal/maximum qualified altitude
Deployment airspeed/Mach	Minimum/nominal/maximum qualified conditions
Atmosphere	Density/temperature/wind envelope and atmospheric model
Canopy geometry	Area, diameter/planform, porosity, line length and material
C_d / aerodynamic model	Validated by test or defensible empirical model
Opening shock	Peak load, duration, structural load path and occupant implications
Reefing / staged deployment	Sequence, timing and load limiting
Failure cases	One/multiple canopy loss, line failure, riser failure, asymmetry, spin/coning
Control	Attitude stability, turn/oscillation limits and landing guidance
Terminal condition	Ground contact vertical/lateral speed and attitude criteria

Mandatory gate: No “safe descent” number may be claimed until a validated 6-DOF trajectory model, parachute structural model, deployment analysis and representative dynamic test programme are complete.

13. Landing Legs and Impact Energy Management

Eight landing legs with progressive hydraulic dampers, elastomer pads and viscoelastic seat suspension remain the baseline architecture. The system shall be treated as an energy-management chain rather than a list of components.

The total translational kinetic energy at impact is $E = \frac{1}{2}mv^2$. The engineering model shall allocate this energy across landing-leg stroke, hydraulic dissipation, elastomer deformation, structural deformation, ground interaction and occupant restraint/seat absorption.

- Define allowable vertical, lateral and combined impact velocities.
- Analyze symmetric, asymmetric and single/multiple-leg-failure cases.
- Include attitude error, uneven terrain, hard-point impact and rollover cases.
- Define damper force-displacement and velocity characteristics over temperature and manufacturing tolerance.
- Define stroke limits, bottoming loads, rebound behaviour and post-impact stability.
- Correlate nonlinear FEA and dynamic tests using measured force, displacement, acceleration and strain.

Occupant injury criteria shall be linked to the vehicle acceleration time histories. A low peak force alone does not establish acceptable injury risk; pulse duration, direction, seat/restraint interaction and human tolerance criteria must be included.

14. Occupant Protection, Seats, Restraints and Egress

- Seats and restraints shall maintain occupant position during normal operation, separation, parachute opening and impact.
- Define occupant anthropometric range and accommodation envelope.
- Restraints shall be compatible with emergency egress and shall not create unacceptable release failure modes.
- Seat energy absorption shall be coordinated with landing-leg and floor dynamics.
- Provide independent or protected emergency lighting and egress marking.
- Demonstrate egress after deformation, inversion, smoke, water ingress and loss of electrical power.
- Define rescue access points and external emergency markings.

The three nominal exits — two side doors and one roof hatch — shall be evaluated for blocked-door, structural deformation and post-water-event conditions. Egress time shall be established by test for the defined occupant population and scenario.

15. Life Support, Atmosphere and Thermal Control

The 72-hour life-support target applies to seven occupants. The final system must close the atmosphere, consumables, power and thermal balances rather than relying on a nominal equipment list.

Balance	Required closure
Oxygen	Consumption rate, storage/generation capacity, reserve and failure modes
CO2	Generation rate, removal capacity, breakthrough and replacement/regeneration
Humidity	Moisture generation, condensation control and equipment tolerance
Ventilation	Flow distribution, fan redundancy, dead zones and smoke mode
Thermal	Occupant + equipment heat rejection for hot/cold environments
Water	Potable and emergency reserve quantities
Food	72 h requirement and packaging/thermal control
Contaminants	Battery/pyro/fire off-gassing and toxic products
Pressure/leak	Cabin pressure strategy, leakage rate and emergency response
Power	Electrical energy required for the complete survival chain

Life-support mass, electrical demand and heat rejection shall be included in the controlled mass and energy budgets.

16. Fire, Smoke and Thermal Protection

Fire protection shall be hazard-based. The capsule contains stored energy, electrical systems, batteries, pyrotechnic devices, hydraulic fluid and potentially high-temperature host environments.

- Identify ignition sources and credible fire scenarios.
- Provide compartmentation or fire barriers where required by hazard analysis.
- Control battery thermal runaway propagation and toxic-gas exposure.
- Protect critical power and separation wiring from common fire/thermal damage.
- Define smoke detection, ventilation isolation and occupant warning.
- Select extinguishing agents compatible with electrical, composite, battery and enclosed-volume hazards.
- Qualify materials for flame, smoke and toxicity requirements appropriate to the intended certification basis.
- Analyze external fire, post-crash fire and water-entry fire cases.

17. Flotation, Stability, Flooding and Self-Righting

The maritime survival function requires quantitative hydrostatic and dynamic analysis. A nominal statement that the capsule floats or self-rights is insufficient.

- Define fully loaded displacement and reserve buoyancy.
- Define watertight volume, flooding rate and allowable damage/flooding cases.
- Calculate freeboard and stability for all relevant mass/CG states.
- Produce righting-arm curves and identify positive-stability range.
- Analyze asymmetric flooding and damaged buoyancy chambers.
- Define self-righting mechanism, energy source, initiation conditions and failure cases.
- Analyze water impact, capsize, rollover and re-entry into the water.
- Verify occupant egress after inversion and partial flooding.
- Define rescue interface, visual aids, acoustic aids and location equipment.

The 406 MHz COSPAS-SARSAT, GNSS and satellite-messaging architecture shall be qualified for antenna visibility, power availability and post-crash orientation constraints.

18. Electrical Power, Software and Common-Cause Control

Redundant processors or duplicated electronics are not automatically independent. Independence must be demonstrated at the architecture level.

- Provide separated power sources or sufficiently independent power paths for safety-critical channels.
- Separate critical wiring routes where a single damage event could defeat both channels.
- Use independent sensors where common-cause sensing failure is credible.
- Control common software libraries, compilers, configuration data and shared services.
- Define watchdogs, safe states, fault containment and deterministic recovery.
- Protect critical functions against EMI/EMC, lightning, static discharge and transient events as applicable.
- Perform software hazard analysis and requirements-based verification for safety-critical logic.
- Perform fault injection at sensor, processor, bus, power and actuator levels.

A common-cause analysis shall explicitly examine fire, flooding, impact, connector damage, software corruption, power loss, thermal runaway and environmental extremes.

19. Pyrotechnic, Thermal-Battery, Pressure-Vessel and Stored-Energy Safety

Stored-energy devices require independent hazard controls. The system shall define safe-arm logic, physical barriers, accidental-initiation resistance, maintenance controls and disposal procedures.

- Pyrotechnic devices: initiation isolation, misfire handling, adjacent-hardware protection and post-event safe state.
- Thermal batteries: activation control, thermal propagation, storage life and end-of-life behaviour.
- Hydraulic/pneumatic stores: pressure-vessel qualification, relief devices, leakage and burst containment.
- Cold-gas fallback: pressure, regulator, valve, thermal and discharge hazards.
- All stored-energy items: transport, maintenance, inspection and fault-containment requirements.

20. Environmental Qualification

The capsule and safety-critical subsystems shall be qualified against the environmental envelope appropriate to the intended host and mission.

Environment	Required assessment
Temperature	Operating, survival, thermal cycling and material-property effects
Pressure/altitude	Equipment operation, cabin behaviour and parachute-relevant atmosphere
Humidity/water	Condensation, corrosion, ingress and insulation effects
Vibration/shock	Host vibration, separation shock, deployment and landing shock
EMI/EMC	Radiated/conducted susceptibility and emissions
Lightning/static	As applicable to host and operational concept
Salt/fog/corrosion	Required for maritime and exposed configurations
UV/weathering	External materials and seals where applicable
Fire/smoke	Material and system-level hazard qualification
Aging	Storage life, fatigue, seal degradation and maintenance intervals

21. Host-Platform Interface Control

The 22,000 kg capsule imposes major host-level feasibility constraints. Each host class requires a dedicated interface control document and feasibility review.

Host class	Mandatory analyses
Helicopter	Gross mass/CG, rotorcraft performance, structural attachment, vibration, aerodynamic interference, separation trajectory, autorotation/emergency states
Fixed-wing / jet	Mass/CG, structural attachment, drag, aeroelastic effects, separation trajectory, high-speed dynamic pressure, flutter/loads and emergency flight states
Maritime / ship	Deck/launch interface, sea-state loads, water impact, buoyancy, crane/handling loads, corrosion and recovery logistics

A host platform shall not be declared compatible merely because its nominal payload capacity exceeds capsule mass. The full mission and emergency load cases must close.

22. Variant and Family Architecture

Three primary variants are controlled: helicopter, jet/fixed-wing and maritime/ship. The common capsule core should be retained only where the physical and certification interfaces genuinely remain common.

- Maintain a common core requirements set for survival functions.
- Create variant-specific interface control documents, aerodynamic models, load cases and qualification plans.
- Do not alter the pure-survival baseline to accommodate a future propulsion or hover concept without a formal configuration change.
- Any variant-specific subsystem that can create a common-cause hazard to the common core shall be isolated by architecture and verified independently.

23. Scalability Rules

Scalability shall preserve the engineering logic rather than simply scale dimensions linearly.

- Mass scaling must be accompanied by parachute area, structural load, impact energy, flotation and host-interface re-analysis.
- Occupancy changes require life-support, seating, egress, mass-properties and emergency-procedure updates.
- Diameter/height changes require new structural, aerodynamic, stability and manufacturing substantiation.
- Changing the number of landing legs requires a new load-sharing and failure-case analysis.
- Changing parachute count or geometry requires a complete recovery-system requalification.
- Any scale derivative shall receive a new configuration identifier and verification matrix.

24. Verification, Validation and Test Programme

Verification shall use the method appropriate to each requirement. The minimum programme is staged from analysis and component testing through integrated system tests.

Level	Representative work
Requirements analysis	Trace every requirement to an owner, input, output, verification method and acceptance criterion.
Component	Materials, dampers, valves, actuators, batteries, parachute components, sensors and communications.
Subassembly	Leg module, cage joint, separation interface, seat/restraint, flotation module, life-support module.
Integrated ground test	Full capsule functional test with representative mass and instrumentation.
Dynamic test	Drop/impact, separation and recovery-system dynamic testing.
Environmental test	Temperature, vibration, shock, humidity, EMI/EMC and other applicable environments.
Full-system qualification	Representative mission/accident scenarios with controlled instrumentation and safety case.
Post-test inspection	Damage, residual deformation, leaks, wiring, composite condition and functional re-test.

Testing shall be designed to discriminate between competing failure mechanisms, not merely demonstrate nominal operation.

25. Mandatory Safety Analyses

- Functional Hazard Assessment (FHA)
- System and subsystem FMEA/FMECA
- Fault Tree Analysis (FTA)
- Common-Cause Analysis (CCA)
- Preliminary/Functional Hazard and Safety Assessment as required by the selected certification basis
- Structural failure-mode and damage-tolerance analysis
- Parachute failure and deployment analysis
- Impact and occupant-injury analysis
- Fire and thermal hazard analysis
- Battery and stored-energy hazard analysis
- Flotation/flooding/capsize analysis
- Human-factors and egress analysis
- Software and hardware safety analysis for safety-critical functions

The outputs shall be controlled as part of the system safety case and shall be updated whenever architecture, interfaces, mass, CG or operating envelope changes.

26. Human Factors and Operational Procedures

- Define normal, emergency and post-event crew procedures.
- Minimize the number of actions required under high workload.
- Provide clear arming/inhibit status and unambiguous emergency indications.
- Prevent inadvertent activation during maintenance and normal operation.
- Design emergency controls for degraded visibility, smoke, noise, vibration and reduced dexterity.
- Train and test representative crews in emergency detection, manual initiation, egress and post-event survival procedures.
- Define maintenance and inspection tasks that cannot accidentally defeat redundancy.

27. Requirements Traceability and Evidence Database

Every requirement shall have a unique identifier and a trace to source, rationale, design element, verification method, verification result and configuration baseline.

Field	Minimum content
Requirement ID	Stable unique identifier
Source	Document, analysis, hazard, customer or certification basis

Field	Minimum content
Rationale	Why the requirement exists
Owner	Responsible engineering function
Verification method	Analysis / inspection / test / demonstration
Acceptance criterion	Pass/fail condition
Evidence	Report, dataset, drawing, test record or certificate
Configuration	Hardware/software/document version tested
Status	Open / verified / waived / superseded

28. Configuration Management

- Lock geometry, mass ceiling, safety architecture and interface definitions through formal change control.
- Assign unique configuration identifiers to hardware, software, CAD, analyses and test articles.
- Record all changes with reason, impact assessment and re-verification requirement.
- Prevent unapproved engineering changes from entering certification evidence.
- Maintain a master list of assumptions and close or retire each assumption before dependent claims are released.
- Maintain a nonconformance and corrective-action system for test and manufacturing findings.

29. Engineering-to-IP Traceability

The concept may support multiple patent families, but any count is preliminary until a professional prior-art, novelty, inventive-step, claim-scope and freedom-to-operate analysis is completed.

Potential IP cluster	Engineering subject to protect if novel
Layered survival architecture	System-level arrangement and failure-tolerant architecture
Clean-break separation interface	Mechanical/electrical/data/fluid separation interface architecture
Multi-path emergency initiation	Independently controlled initiation and fallback logic
Impact-energy architecture	Integrated leg/seat/structural energy management
Flotation/self-righting	Integrated buoyancy, stability and recovery architecture
Host/capsule interface	Variant-specific interface and separation architecture

IP extraction shall be traceable to implemented engineering features. Patent claims shall not rely on unsupported performance statements.

30. Engineering Maturity Gates

Gate	Minimum exit criteria
G0 — Concept	Mission, architecture, hazards and locked top-level values defined.
G1 — System Requirements	Requirements allocated, interfaces identified, mass/CG framework established.
G2 — Preliminary Design	Loads, energy budgets, parachute sizing basis, flotation and life-support balances established.
G3 — Detailed Design	Manufacturing definition, drawings/models, analysis and failure cases substantially closed.
G4 — Qualification Readiness	Verification procedures, instrumentation, test articles and safety plans approved.
G5 — Qualification	Required tests passed and evidence correlated with analysis.
G6 — Certification / Operational Release	Applicable certification basis satisfied and configuration formally released.

A gate is not passed because a document exists; it is passed when the defined evidence and acceptance criteria are satisfied.

31. Mandatory Engineering Work Packages Before Performance Claims

- Close the 22,000 kg mass budget with explicit hardware, tolerance, growth and headroom.
- Produce a controlled 3-axis CG and inertia database.
- Complete parachute aerodynamic/trajectory sizing and dynamic deployment analysis.
- Complete impact energy allocation and occupant injury analysis.
- Complete nonlinear structural analysis and test correlation.
- Complete six-layer independence proof using FMEA/FTA/CCA and fault injection.
- Complete host-specific load/separation feasibility for each intended host class.
- Complete flotation hydrostatics, righting and flooding analysis.
- Close 72-hour life-support mass, energy, thermal and consumables balances.
- Complete fire/smoke/battery/stored-energy hazard analysis.
- Complete electrical/software common-cause and fault-injection analysis.
- Build the requirements verification matrix and evidence database.
- Define manufacturing, inspection, repair and maintenance controls for safety-critical structure and systems.

Release rule: Until these work packages are sufficiently closed, Document 00 may support concept definition and IP strategy, but it shall not be represented as a fully qualified aircraft/ship safety system or certification-ready product.

32. Controlled Baseline Summary

Parameter	Version 3.0 controlled position
External diameter	5.80 m
Overall height	3.80 m
Occupancy	7 persons
Maximum operational mass / hard ceiling	22,000 kg
Primary structure	Continuous CFRP roll-cage + titanium nodes + continuous floor
Primary tube class	15–18 cm class, final sizing by analysis
Primary structural SF target	≥2.8, with defined limit/ultimate basis required
Landing legs	8, progressive hydraulic + elastomer energy management
Parachutes	8 primary + 8 auxiliary/stabilizer concept; sizing and qualification open
Life support	72 h target for 7 occupants
Location	406 MHz COSPAS-SARSAT + GNSS + satellite messaging
Exits	2 side doors + 1 roof hatch
Safety layers	6
Variants	Helicopter / fixed-wing-jet / maritime-ship
Certification status	Not certified; verification and qualification required

This table is the controlled system-level summary. Lower-level documents may refine values only through configuration control and traceability.

33. Corrected Deficiency Register — Version 1.6 to Version 3.0

Deficiency	Correction in Version 3.0
Mass budget did not close to 22,000 kg	Explicitly quantified residual/headroom and imposed mass-control rules.
"Independent" safety layers were asserted qualitatively	Independence is now tied to FMEA/FTA/CCA, physical/electrical/software separation and fault injection.
Parachute sizing incomplete	Required variables, equations, failure cases and qualification gates are defined.
Impact survival lacked energy accounting	Energy equation, allocation chain, combined/asymmetric cases and test correlation are required.
CG was a nominal point	Full configuration envelope and inertia control added.
Structural SF lacked load-basis definition	Limit/ultimate loads, allowables, damage state and correlation are mandatory.
Host compatibility was too general	Dedicated host-class interface and feasibility analysis required.
Flotation was qualitative	Hydrostatics, righting, flooding, damage and egress analysis required.
Life support was equipment-oriented	Atmosphere, thermal, consumable and power balances required.
Fire protection was generic	Hazard-based fire, battery, pyro and toxic-atmosphere analysis required.
Electrical redundancy could hide common causes	CCA, power/sensor/software separation and fault injection required.
Egress was under-defined	Abnormal, inverted, smoke, water and power-loss egress testing required.
Patent-family count could be overinterpreted	IP count is explicitly preliminary and subject to professional legal analysis.
Version could be mistaken for certification	Status and release gates explicitly separate baseline definition from qualification/certification.

34. Final Engineering Position

Version 3.0 is the corrected system-level engineering baseline for the Ultra-Safe Central Capsule concept. It is materially more rigorous than the earlier baseline because it separates locked values from unresolved engineering, closes the mass-accounting logic, defines the evidence required for redundancy and survival claims, and establishes a verification-gated path to detailed design and qualification.

The architecture remains technically ambitious. The principal technical risks are not the existence of the six-layer concept itself, but the quantitative substantiation of a 22,000 kg capsule: host compatibility, parachute deployment and controllability, structural load paths, impact energy absorption, occupant injury limits, mass/CG closure, flotation stability, 72-hour life support, stored-energy hazards, and common-cause failure control.

Final release statement: This document is suitable as a controlled system engineering baseline and as a source for detailed engineering work packages and IP extraction. It is not evidence that the capsule has already demonstrated the claimed survival performance. Any statement of certified performance shall be made only after the applicable analyses, tests and qualification evidence are completed and approved.

END OF DOCUMENT 00 — VERSION 3.0